Cirrus SR20 G3 / G6 (SR20)

Continental IO-360-ES, 200 hp (G3) / Lycoming IO-390-C3B6, 215 hp (G6) · 4 (G6: 5) seats · Typical SR20 G3 (IO-360-ES, Avidyne or Perspective) and G6 (IO-390, Perspective+) POH procedures; speeds are G3 figures, G6 differs by a few knots.

V-speeds & limits

Vr	**65–70 KIAS (flaps 50%)**
Vx	**81 KIAS**
Vy	**96 KIAS**
Best glide	**87–88 KIAS**
Vfe	**119 KIAS (50%) / 104 KIAS (100%)**
Va	**130 KIAS (3,050 lb)**
Vno	**163 KIAS**
Vne	**200 KIAS**
Vpd (CAPS max demonstrated)	**133 KIAS**
Vs / Vso	**65 / 56 KIAS**
Vref	**75–80 KIAS (flaps 100%)**

Normal procedures

Before Start

- Preflight inspection — **COMPLETE (fuel drains, oil 6–8 qt, pitot cover removed)**
- CAPS handle safety pin — **REMOVED (handle cover closed, pin stowed)**
- Passenger briefing — **COMPLETE (belts, doors, CAPS location & use, exits)**
- Seats, belts, harnesses — **ADJUST & LOCK**
- Doors — **LATCHED (both handles)**
- Brakes — **TEST & SET (parking brake)**
- Fuel selector — **FULLER TANK (LEFT or RIGHT)**
- Power lever — **IDLE**
- Mixture — **CUTOFF**
- Alternate induction air — **CLOSED (pushed in)**
- Bat 2 — **ON – check volts 23–25 V**
- Bat 1 — **ON**
- Avionics — **OFF (Perspective: avionics master OFF)**
- Annunciators / CAS — **CHECK (no red)**
- Fuel quantity — **CHECK (vs visual check)**
- Flaps — **CHECK operation (50%, 100%), set UP**
- Strobes / nav lights — **ON**

Engine Start

Propeller area	**CLEAR**
Power lever	**FULL FORWARD**
Mixture	**FULL RICH**
Fuel pump	**PRIME (hold 2–3 s cold, none when hot), then BOOST**
Power lever	**OPEN 1/4 in**
Ignition	**START (max 20 s, release when engine fires)**
Power lever	**1,000 RPM**
Oil pressure	**CHECK within 30 s (green 30–60 psi)**
Alt 1 / Alt 2	**ON – check charging, no ALT 1 / ALT 2 caution**
Fuel pump	**BOOST (leave on until after takeoff)**
Avionics	**ON**
Engine instruments	**CHECK**
Pitot heat	**TEST then OFF**

Before Taxi

Flight instruments / PFD	**CHECK & SET (altimeter, HSI, CDI)**
Radios / transponder	**SET**
Trim	**SET for takeoff (neutral mark)**
Flaps	**50%**
Parking brake	**RELEASE**
Brakes	**CHECK immediately after moving**

Nosewheel is free-castoring: steer with differential braking at low speed. Avoid prolonged brake drag – brake overheat is a fire risk.

HSI / turn indication	**CHECK during turns**

Before Takeoff

Parking brake	**SET**
Doors	**LATCHED**
CAPS handle	**VERIFY pin removed, cover closed**
Seat belts / harnesses	**SECURE**
Flight controls	**FREE & CORRECT**
Fuel selector	**FULLER TANK**
Fuel pump	**BOOST**
Mixture	**FULL RICH (lean for field elevation above 3,000 ft)**
Power lever	**1,700 RPM**
Alt 1 / Alt 2	**CHECK positive amps, Ess Bus / Main Bus volts green**
Magnetos	**CHECK L / R (max drop 150 RPM, max diff 75 RPM)**
Engine instruments	**CHECK green (oil temp min 100°F for takeoff)**
Power lever	**IDLE check (600–800 RPM), then 1,000 RPM**
Flaps	**50% (check indication)**
Trim	**SET takeoff**
Autopilot	**CHECK / disconnect**
Pitot heat / lights	**AS REQUIRED**
Transponder	**ALT**
Takeoff briefing	**COMPLETE (Vr 65–70, Vy 96, abort point, CAPS available above approx 600 ft AGL)**

After Takeoff / Climb

Airspeed	**Vy 96 KIAS (obstacles: Vx 81 KIAS)**
Flaps	**UP (above 80 KIAS, positive rate)**
Fuel pump	**OFF above 1,000 ft AGL (BOOST if fuel flow fluctuates)**
Power	**FULL (cruise climb 110–120 KIAS)**
Mixture	**FULL RICH (lean per fuel flow placard above 3,000 ft)**
Engine instruments	**CHECK (CHT max 460°F, oil temp max 240°F)**

Cruise

Power	**SET (typ. 55–75%, 2,500–2,700 RPM)**
Mixture	**LEAN (per power table / lean-assist, 50°F rich of peak or LOP as approved)**
Fuel pump	**OFF (BOOST if fuel flow fluctuates or hot/high)**
Fuel selector	**SWITCH tanks every 30 min, keep within 10 gal imbalance**
Fuel quantity / totalizer	**CHECK vs flight plan**
Engine instruments	**CHECK**

Descent

Altimeter	**SET**
Mixture	**ENRICH as altitude decreases**
Power	**REDUCE gradually (keep CHT above 250°F)**
Fuel selector	**FULLER TANK**
Approach briefing	**COMPLETE (ATIS, approach, speeds)**

Before Landing

Seats / belts	**SECURE**
Fuel selector	**FULLER TANK**
Fuel pump	**BOOST**
Mixture	**FULL RICH (as required for altitude)**
Flaps	**50% below 119 KIAS, 100% below 104 KIAS**
Airspeed	**FINAL 75–80 KIAS (100%), 80–85 KIAS (50%)**
Autopilot	**OFF**
Landing light	**ON**

After Landing

Flaps	**UP**
Fuel pump	**OFF**
Pitot heat	**OFF**
Transponder	**STBY / as required**
Lights	**AS REQUIRED**
Power lever	**1,000 RPM taxi**

Shutdown

Parking brake ..... **SET**

Power lever ..... **IDLE, 1 min cool-down**

Avionics ..... **OFF**

Mixture ..... **CUTOFF**

Ignition ..... **OFF (after prop stops), key removed**

Alt 1 / Alt 2 ..... **OFF**

Bat 1 / Bat 2 ..... **OFF**

CAPS handle safety pin ..... **INSERT**

Fuel selector ..... **LEFT or RIGHT (leave on a tank)**

Pitot cover / chocks / tie-downs ..... **INSTALL**

Abnormal procedures

ALT 1 or ALT 2 Failure

ALT 1 / ALT 2 caution, Main or Ess Bus amps zero

ALT 1 / ALT 2 circuit breaker (affected) ..... **CHECK IN**

Alt 1 / Alt 2 (affected) ..... **OFF, then ON (reset once)**

▶ Alternator restored?

Yes:

Electrical load ..... **MONITOR**

No – ALT 1 failed:

Alt 1 ..... **OFF**

Non-essential loads ..... **OFF (Alt 2 supplies approx 20 A to the essential bus only)**

Bat 1 ..... **MONITOR – powers main bus until depleted (approx 30 min)**

Deferred: ALT 1 failed – main bus on battery, limited endurance; flaps once on final, land as soon as practical. ..... **NOTED**

No – ALT 2 failed:

Alt 2 ..... **OFF**

Flight ..... **CONTINUE – Alt 1 supplies both buses**

Deferred: ALT 2 failed – no electrical redundancy, avoid IMC / night if practical, land as soon as practical. ..... **NOTED**

Low Volts / Both Alternators Failed

LOW VOLTS warning on Main and Ess bus

Alt 1 / Alt 2 ..... **OFF then ON (reset once)**

▶ Voltage recovered?

Yes:

Electrical load ..... **MONITOR**

No:

Alt 1 / Alt 2 ..... **OFF**

Bat 1 ..... **OFF (save Bat 2 for essential bus)**

Non-essential loads ..... **OFF**

Bat 2 powers the essential bus (PFD, standby attitude, COM1, NAV1) for approx 30 min. Flaps and trim are on the main bus – turn Bat 1 ON briefly for flaps on final.

Deferred: Both alternators failed – battery only, flaps once on final with Bat 1 ON, land as soon as possible. ..... **NOTED**

Low Oil Pressure / High Oil Temperature

OIL PRESS warning (below 30 psi, 10 psi idle), oil temperature above 240°F or CHT above 460°F

Oil pressure / temperature / CHT ..... **CHECK all, cross-check**

▶ **Which indication?**

Low pressure:

▶ **Oil temperature rising or pressure below 10 psi?**

Yes:

Power ..... **REDUCE to minimum required**

Landing site ..... **SELECT, expect engine failure; CAPS if no safe site**

Deferred: Low oil pressure – land as soon as possible, engine failure possible. ..... **NOTED**

No – temperature normal:

Flight ..... **CONTINUE, monitor closely**

Deferred: Suspect oil pressure indication – land as soon as practical. ..... **NOTED**

High temperature:

Mixture ..... **ENRICH**

Airspeed ..... **INCREASE (lower nose)**

Power ..... **REDUCE**

▶ **Temperature returning to green?**

Yes:

Flight ..... **CONTINUE, monitor**

No:

Power ..... **MINIMUM practical**

Deferred: Engine overheating – land as soon as practical, expect possible power loss. ..... **NOTED**

Rough Running / Loss of Power

Fuel pump ..... **BOOST**

Mixture ..... **ADJUST (richer, then leaner)**

Fuel selector ..... **SWITCH tanks**

Alternate induction air ..... **OPEN (induction icing or blockage)**

Ignition ..... **CHECK L / R individually, BOTH**

▶ **Engine running smoothly?**

Yes:

Configuration ..... **KEEP the working combination**

Deferred: Engine ran rough – land as soon as practical, maintenance check. ..... **NOTED**

No:

Power ..... **REDUCE to smoothest setting**

Land ..... **AS SOON AS POSSIBLE (prepare for engine failure / CAPS)**

Low Fuel Flow / Fuel Flow Fluctuation

Fuel flow fluctuating, FUEL PRESS or low fuel caution

Fuel pump ..... **BOOST**

Fuel selector ..... **SWITCH to fuller tank**

Mixture ..... **FULL RICH**

Fuel quantity ..... **CHECK both tanks**

▶ **Fuel flow steady?**

Yes:

Fuel pump ..... **LEAVE ON BOOST**

Deferred: Fuel pump on BOOST continuously – engine-driven pump or vapour suspect. ..... **NOTED**

No:

Land ..... **AS SOON AS POSSIBLE, expect engine failure**

Flap Failure

FLAPS circuit breaker ..... **CHECK / RESET once**

Flap switch ..... **CYCLE (UP / 50% / 100%)**

▶ Flaps operable?

Yes:

Flap setting ..... **AS REQUIRED**

No:

Approach ..... **FLAPS UP 85–90 KIAS / 50% 80–85 KIAS, flat approach**

Deferred: Flaps inoperative – flaps-up landing, increased approach speed and landing distance. ..... **NOTED**

Door Open in Flight

Airspeed ..... **REDUCE below 119 KIAS**

Door ..... **DO NOT attempt to close – hold shut if practical**

A door open in flight is noisy and causes buffet; the aircraft remains controllable.

Loose objects / belts ..... **SECURE**

Deferred: Door open – land as soon as practical, keep speed below 119 KIAS. ..... **NOTED**

PFD / AHRS / ADC Failure

Display backup (red button) ..... **PRESS – reversionary mode on MFD**

Standby instruments ..... **USE (airspeed, attitude, altimeter, compass)**

Autopilot ..... **OFF if AHRS failed**

PFD / AHRS circuit breakers ..... **CHECK**

▶ Flight in IMC?

Yes:

ATC ..... **ADVISE, request radar assistance / nearest VMC**

Deferred: Primary flight display failed – partial panel, request vectors, avoid IMC. ..... **NOTED**

No:

Deferred: PFD / AHRS failed – standby instruments only. ..... **NOTED**

Static Source Blocked / Erroneous Airspeed

Pitot heat ..... **ON**

Alternate static source ..... **OPEN (valve under panel, left of pilot)**

Airspeed / altitude ..... **CROSS-CHECK with GPS ground speed and standby altimeter**

Attitude / power ..... **FLY known pitch and power settings**

Deferred: Pitot-static unreliable – fly attitude and power, add margin to approach speed. ..... **NOTED**

Emergency procedures

CAPS Deployment

Loss of control, mid-air collision, structural failure, pilot incapacitation, or engine failure where a safe forced landing is not assured

Airspeed (memory)	**MINIMISE (below 133 KIAS, as slow as practical)**
CAPS handle (memory)	**PULL straight down, both hands, full stroke (approx 45 lb)**
Mixture (memory)	**CUTOFF**
Fuel selector (memory)	**OFF**
Bat 1 / Bat 2 / Alt 1 / Alt 2 (memory)	**OFF**
Seat belts / harnesses (memory)	**TIGHT**
Brace (memory)	**ASSUME emergency landing body position (arms crossed, hands on shoulders)**

CAPS is the primary response to loss of control, mid-air collision, structural failure, pilot incapacitation and engine failure over terrain where a safe forced landing is not assured. Recommended decision altitude 2,000 ft AGL; minimum demonstrated deployment from a loss of control 920 ft AGL, from level flight approx 400 ft AGL. Below the minimum, deploy anyway – it is still the best option.

Ignition / fuel pump	**OFF (if time permits)**
ELT	**ON (if time permits)**
Descent	**EXPECT approx 1,700 ft/min, 8 s to full canopy, hard landing (approx 3 g)**
After touchdown	**EVACUATE upwind if canopy drags, aircraft may be dragged by wind**

Engine Failure During Takeoff Roll

Power lever (memory)	**IDLE**
Brakes (memory)	**APPLY (differential for directional control)**
Flaps	**UP**
Mixture	**CUTOFF**
Fuel selector	**OFF**
Ignition	**OFF**
Bat 1 / Bat 2 / Alt 1 / Alt 2	**OFF**

Engine Failure after Takeoff

Airborne, insufficient runway remaining

LAND IMMEDIATELY

Airspeed (memory)	**BEST GLIDE or landing speed (80 KIAS flaps 50%)**
Mixture (memory)	**FULL RICH**
Fuel selector (memory)	**SWITCH tanks**
Fuel pump (memory)	**BOOST**
Alternate induction air (memory)	**OPEN**
Ignition (memory)	**BOTH (check)**

Below approx 600 ft AGL land straight ahead with only small heading changes – do not turn back. Above approx 600 ft AGL CAPS deployment is an option if a safe landing area is not available.

▶ Safe landing area ahead?

Yes – land ahead:

Flaps	**100% when landing assured**
Mixture	**CUTOFF**
Fuel selector	**OFF**
Ignition	**OFF**
Bat 1 / Bat 2 / Alt 1 / Alt 2	**OFF**
Doors	**UNLATCH before touchdown**

No – CAPS:

Deploy CAPS – see CAPS Deployment procedure.

Engine Failure in Flight

Loss of power with altitude available

LAND AS SOON AS POSSIBLE

Airspeed (memory)	**88 KIAS (best glide)**
Mixture (memory)	**FULL RICH**
Fuel selector (memory)	**SWITCH tanks**
Fuel pump (memory)	**BOOST**
Alternate induction air (memory)	**OPEN**
Ignition (memory)	**BOTH (cycle L / R / BOTH)**
Landing site	**SELECT, turn towards it**

▶ Restart attempt practicable?

Yes – restart:

Power lever	**1/2 in open**
Mixture	**FULL RICH, then lean slowly**
Ignition	**START if propeller stopped**

▶ Engine restarted?

Yes:

Power	**AS REQUIRED (minimum needed)**
Engine instruments	**MONITOR**
Deferred: Engine restarted after in-flight failure – land as soon as possible, suspect fuel or induction problem.	**NOTED**

No:

Continue with forced landing or CAPS.

No – forced landing / CAPS:

▶ Safe forced landing assured?

Yes – forced landing:

Mayday / 7700	**TRANSMIT**
Seats / belts	**SECURE**
Flaps	**100% when landing assured**
Airspeed	**75–80 KIAS on final**
Mixture	**CUTOFF**
Fuel selector	**OFF**
Ignition	**OFF**
Bat 1 / Bat 2 / Alt 1 / Alt 2	**OFF**
Doors	**UNLATCH before touchdown**

No – deploy CAPS:

Deploy CAPS before descending through 2,000 ft AGL – see CAPS Deployment procedure.

Engine Fire in Flight

LAND IMMEDIATELY

- Mixture (memory) ..... **CUTOFF**
- Fuel pump (memory) ..... **OFF**
- Fuel selector (memory) ..... **OFF**
- Power lever (memory) ..... **IDLE**
- Ignition (memory) ..... **OFF**
- Cabin heat / defrost (memory) ..... **OFF**
- Airspeed ..... **INCREASE (sideslip to keep flames from cabin)**

▶ Fire extinguished and landing area within gliding range?

Yes:

- Forced landing ..... **EXECUTE (do not attempt restart)**

No:

Deploy CAPS – see CAPS Deployment procedure.

Engine Fire During Start

- Mixture (memory) ..... **CUTOFF**
- Fuel pump (memory) ..... **OFF**
- Fuel selector (memory) ..... **OFF**
- Power lever (memory) ..... **FULL FORWARD**
- Ignition (memory) ..... **CONTINUE cranking (draw fire into engine)**
- Ignition / Bat / Alt ..... **OFF when fire out**
- Fire extinguisher ..... **OBTAIN**
- Aircraft ..... **EVACUATE**

Cabin / Electrical Fire

LAND AS SOON AS POSSIBLE

- Bat 1 / Bat 2 (memory) ..... **OFF**
- Alt 1 / Alt 2 (memory) ..... **OFF**
- Avionics (memory) ..... **OFF**
- Cabin heat / fresh air ..... **CLOSE**
- Fire extinguisher ..... **ACTIVATE**

After discharge, open fresh air vents / crack a door to ventilate once fire is out.

▶ Fire out and electrical power required?

Yes:

- Circuit breakers ..... **CHECK – do not reset tripped breakers**
- Bat 2 / Alt 2 ..... **ON (essential bus first)**
- Bat 1 / Alt 1 / avionics ..... **ON one at a time, check for smoke**
- Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable. ..... **NOTED**

No:

Continue with electrics off. Standby instruments and magnetic compass only; no flaps, no electric trim.

- Deferred: All electrics off – flaps-up landing approx 90 KIAS, no radios, no trim. ..... **NOTED**

Emergency Descent

- Power lever (memory) ..... **IDLE**
- Mixture (memory) ..... **AS REQUIRED**
- Flaps (memory) ..... **UP**
- Airspeed (memory) ..... **163 KIAS smooth air / 130 KIAS turbulence**

Inadvertent Spin Entry

CAPS handle (memory) ..... **PULL immediately**

The SR20 is not approved for spins and no spin recovery has been demonstrated – CAPS deployment is the only approved recovery. Minimum demonstrated deployment 920 ft AGL.

Power lever ..... **IDLE**

Controls ..... **NEUTRAL**

After deployment ..... **CONTINUE CAPS Deployment procedure**

Prototype content. Verify against the operator's approved checklist and the POH/AFM before any use. AeroE Checklist version 1.7 (build 0d080c7b) · 2026-11-07